

YUHAO HE

(+86)17765960884 (WeChat:13006750206)

3250004430@student.must.edu.mo

[Google Scholar](#) | [Github](#)

EDUCATION

Macau University of Science and Technology, Macau, China PhD in Artificial Intelligence Faculty of Innovation Engineering	Sept. 2025 - present GPA: 3.85/4
Macau University of Science and Technology, Macau, China M.S. in Intelligent Technology Faculty of Innovation Engineering	Sept. 2023 - Jun. 2025 GPA: 3.71/4
Foshan University, Guangdong, China B.S. in Mathematics and Applied Mathematics(AI Innovation Class) Department of Mathematics and Big Data	Sept. 2019 - Jun. 2023 GPA: 86.8/100 Rank: Top 5/72

RESEARCH INTERESTS

1. Adversarial attack; 2. Poisoning attack 3. AIGC safety;

SELECTED RESEARCH EXPERIENCE

Privacy Shield for User Images in AIGC Applications <i>Macau University of Science and Technology, Advisor: Jinyu Tian</i>	Present
• We focused on developing a cross-models universal image perturbation method to safeguard user privacy by adding subtle perturbations, preventing image abuse in AIGC based edit.	
Model Adversarial Robustness Weakening Attack <i>Macau University of Science and Technology, Advisor: Jinyu Tian</i>	Sept. 2023 - May 2024
• We proposed a novel high-threat attack that ensures normal training performance but induces extreme model fragility to minor perturbations and even random noise.	
AI-Powered COVID-19 Diagnosis System <i>Foshan University, Advisor: Xianwei Zheng</i>	Dec. 2021 - Jan. 2023
• We developed an innovative neural network for COVID-19 diagnosis from cough sounds, achieved 96% accuracy and 91% recall on public datasets.	

PUBLICATIONS & MANUSCRIPTS

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- **Yuhao He**, Jinyu Tian, Xianwei Zheng, Li Dong, Yuanman Li and Jiantao Zhou, Deferred Poisoning: Making the Model More Vulnerable via Hessian Singularization, *In Proceedings of the 40th AAAI Conference on Artificial Intelligence*, 2026. (**CCF-A**)
 - **Yuhao He**, Jinyu Tian, Haiwei Wu and Jianqing Li, Structure Disruption: Subverting Malicious Diffusion-Based Inpainting via Self-Attention Query Perturbation, *arXiv:2505.19425*, 2025.
 - **Yuhao He**, Xianwei Zheng and Qing Miao, TFA-CLSTMNN: Novel convolutional network for sound-based diagnosis of COVID-19, *International Journal of Wavelets, Multiresolution and Information Processing* 21.03 (2023): 2250058, 2023. (**SCI**)

WORK EXPERIENCE

ZhuHai UM Science & Technology Research Institue

Jul. 2023 - Sep. 2023

Research Assistant

- Led the development of a model replacement platform leveraging SAM for precise clothing segmentation and Stable Diffusion for inpainting, allowing for effortless model substitution with zero additional cost.
- Integrated the solution into an API and deployed it on a web server using Flask, providing a seamless client-side model customization experience. This project has been successfully delivered and is now in active client use.

JOYY(Guangzhou)

Feb. 2023 - Jun. 2023

Algorithm Intern

- Responsible for the research and development and maintenance of risk control algorithms for social products.
- In charge of daily monitoring of market data, identifying vulnerabilities in current risk control rules, and performing maintenance. During my tenure, I successfully reduced the false negative rate of the risk control algorithm from 12% to 5%.

Foshan Dayan data Technology Co., LTD

Jun. 2021 - Aug. 2021

Algorithm and Development Intern

- Contributed to a deep learning project for strawberry disease and pest identification, which involved curating a dataset of over 30,000 images across 29 categories. Employed SE-ResNet50($32 \times 4d$) model, image augmentation, and transfer learning to achieve 89% accuracy on 14 categories.
- Led the development of a WeChat mini-program, which secured software copyright. This project also applied a patent and was successfully implemented in local plantations in the Foshan area.

PATENTS & COPYRIGHTS

- **[Patent]** Chuntao Jiang, Guanshou Gan, Xuncan Xiao, **Yuhao He**, et al., “A Method and Device for Pest and Disease Identification Based on a Residual Network,” Chinese Patent Application No. 202210536421.0, Foshan University, Publication No. CN115035511A, Sept. 9, 2022. (Notice to Complete Registration Formalities issued Sept. 21, 2026.)
- **[Copyright]** Foshan Dayan Data Technology Co., Ltd., **Yuhao He**, et al., “Strawberry Pest and Disease Identification Mini Program V1.0,” Software Copyright Registration No. 2022SRA003183, National Copyright Administration of the People’s Republic of China, Nov. 7, 2021.

SELECTED HONORS & AWARDS

- Foshan University Outstanding Undergraduate Graduates 2023
- Foshan Specialty Medical Catheters Innovation and Entrepreneurship Scholarship Recipients 2023
- Huazhong Agricultural University Outstanding Summer Camp Member 2022
- **National Scholarship for Undergraduates** 2021
- National Second Prize: National College Students Mathematical Modeling Competition 2021
- National Second Prize: Guangdong Hong Kong Macao Mathematical & Financial Modeling 2020

ADDITIONAL INFORMATION

- Language: Mandarin(Native), Cantonese(Native), English(CET-6)